

NUSRAT JAHAN BABLY

Computer Science Graduate | Teaching & Research | Software Engineering & Data Science

+8801828624088

nbably4088@gmail.com

nusrat-jahan-bably

github.com/nusratbably

Portfolio

Computer Science graduate with strong interests in teaching, research-driven innovation, software engineering, and data science. Passionate about contributing to academic environments that encourage critical thinking, practical learning, and continuous exploration of emerging technologies. Interested in bridging theoretical knowledge with real-world problem solving while pursuing meaningful growth in teaching, research, and higher education.

WORK EXPERIENCE

Grader & Undergraduate Teaching Assistant

United International University (UIU)

Oct 2024 – Feb 2026

Department of CSE & Data Science

UGA

- Assisted undergraduate students in **Object Oriented Programming** and **Data Structures & Algorithms** labs
- Provided debugging support and academic guidance during lab sessions and helped students strengthen problem-solving and programming fundamentals.
- Evaluated assignments, coding tasks, and class assessments with consistency and fairness.
- Collaborated with faculty members in preparing lab materials and ensuring smooth execution of lab activities.
- Supported students who were hesitant to ask questions by providing one-on-one guidance during counselling hours.

- Evaluated assignments and class tests with accuracy and consistency, ensuring fair grading aligned with course objectives.
- Provided constructive feedback to help students improve understanding of core **Electrical Circuit** concepts and theoretical foundations.

Grader

TECHNICAL PROJECTS

BiblioTheca: An Interactive AI-Powered Gamified Library Management System

Tech: React, JavaScript, Spring Boot, Java, H2 Database, REST API, CSS

GitHub

- Developed a full-stack gamified library management system using React and Spring Boot
- Implemented book management, authentication, routing, and REST API features
- Designed an AI-assisted chatbot module and interactive dungeon mini-game system
- Built responsive dark-mode UI with modern user-centric design

Landlytics: Intelligent Land Analytics Platform

Tech: Laravel, PHP, MySQL, Blade, Tailwind CSS, JavaScript, REST APIs

GitHub

- Developed a Laravel-based platform for land management, agricultural analysis, weather monitoring and crop recommendation.
- Built modules to support data-driven land decision-making and agricultural planning.

Medica DB: A Full-Stack Medical Workflow Management System

Tech: React, Node.js, Express, REST APIs, Socket.io, SQL, HTML5 Canvas, JavaScript

GitHub

- Enabled doctors to manage patients, process medical images, collaborate through professional forums, and monitor activities in real time within a responsive interface.

Fuel Theft & Consumption Monitoring System

Arduino, C/C++, Ultrasonic Sensor, Flow Meter, Speed Sensor, Bluetooth Module (HC-05), I2C LCD

ProjectLink

- Built an Arduino-based fuel monitoring system with theft detection and real-time consumption tracking.
- Integrated ultrasonic, flow, and speed sensors to analyze fuel usage and efficiency.
- Enabled Bluetooth data transmission and live status display through an I2C LCD.

RESEARCH EXPERIENCE

Multimodal Speech Reconstruction for Bengali Stroke Patients

Combining Lip Movements and Audio Signals | Final Year Design Project (Thesis)

[Ongoing] | Poster

- Developing a deep learning-based system to reconstruct fluent Bengali speech from lip movements and degraded audio signals of stroke survivors; achieved **Champion** position in the FYDP poster competition.

ShunoBondhu: A Voice Assistive App for Pedal Rickshaw Pullers

Human-Centered Computing Solution Research

[Ongoing]

- Designing a lightweight voice-assisted mobile application that enables non-literate users to interact through simple voice commands for accessing local services, communication support, and everyday assistance.

SmartBoardVision: An Intelligent Classroom Visibility Enhancement System Using Deep Learning

Deep Learning and Digital Image Processing Research

[Ongoing]

- Building an intelligent classroom text visibility pipeline using YOLOv8, EAST, and CRAFT for detection alongside CLAHE and SRGAN for image enhancement.

EDUCATIONAL QUALIFICATIONS

B.Sc. in Computer Science & Engineering (BSCSE) | Graduation year: 20 Feb, 2026

United International University (UIU), Dhaka, Bangladesh

CGPA: 3.90 / 4.00

Higher Secondary Certificate (HSC)

Ispahani Cantonment Public School and College, Cumilla Cantonment

GPA: 4.83

SSC | JSC

GPA: 5 | GPA: 5

SKILLS

Programming Languages: Python, Java, PHP, JavaScript, SQL, C/C++, HTML, CSS.

Frameworks & Tools: Spring Boot, Laravel, Git/GitHub, Maven, MySQL, H2, XAMPP, LaTeX, Jira, Canva, Microsoft Office.

AI & Data Science: Machine Learning, Deep Learning, NLP, TensorFlow, Pandas, NumPy, Matplotlib, Digital Image Processing.

Research & Development: Literature Review, Experimental Design, Data Preprocessing, Prototype Development, Research Writing.

Core Concepts: OOP, DSA, Software Engineering, DBMS, RDBMS, Artificial Intelligence & Machine Learning, Computer Networks & Security, Full-Stack Development, Gamification Design.

Professional Skills : Communication, Classroom Facilitation, Mentoring, Public Speaking, Team Collaboration, Problem Solving, Time Management

ACHIEVEMENTS & ADDITIONAL ACTIVITIES

- Champion** – Final Year Design Project at CSE project show.
- Champion** – Intra University Chess Competition.
- Academic Scholarship Recipient** – Earned 100% (6 times), 50% (4 times), and 25% (2 times) tuition waivers based on academic performance at UIU.
- Champion** – 1st Mirror Inter College National Debate Competition | **Participant** – National High School Programming Contest.(ICT Division)

REFERENCES

Dr. Suman Ahmed

[View LOR]

Dr. Mohammad Nurul Huda

[View LOR]